CYBERSECURITY INTERN REPORT AT SHADOWFOX

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Task Level: Beginner & Intermediate Level
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INTRODUCTION

This report will contain and document all the tasks I performed during my cybersecurity internship at ShadowFox. My main objective about this internship is to gain more knowledge and experience in penetration testing, vulnerability assessment and network security using industry standard tools.

Task Level (Beginner)

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[Task 1] — Find All Open Ports on http://testasp.vulnweb.com/

Attack Name: Port Scanning

Severity: High (Score 7.0 – 8.9)

Reason: HTTP port 80 is considered highly severe because it is unencrypted, making it extremely susceptible to eavesdropping and data interception by malicious actors. This vulnerability can expose sensitive information, such as login credentials and credit card details, when used for web browsing.

Target & Resources Used

- Target: http://testasp.vulnweb.com
- Kali Linux operating system on Parallels Desktop on Mac
- Network Mapping (Nmap)
- Ping tool to get the IP address of the website (ping testasp.vulnweb.com) — run on Mac terminal

Steps to Reproduce

Step 1: Target IP Identification — The website's IP address was identified using the ping command as a precursor to the port scanning process.

Step 2: Port Scanning using Nmap — Using Nmap, a powerful network scanning tool, a port scan was conducted on the identified IP address (44.238.29.244). The goal was to uncover open ports and their associated services.

Results of the port scan:

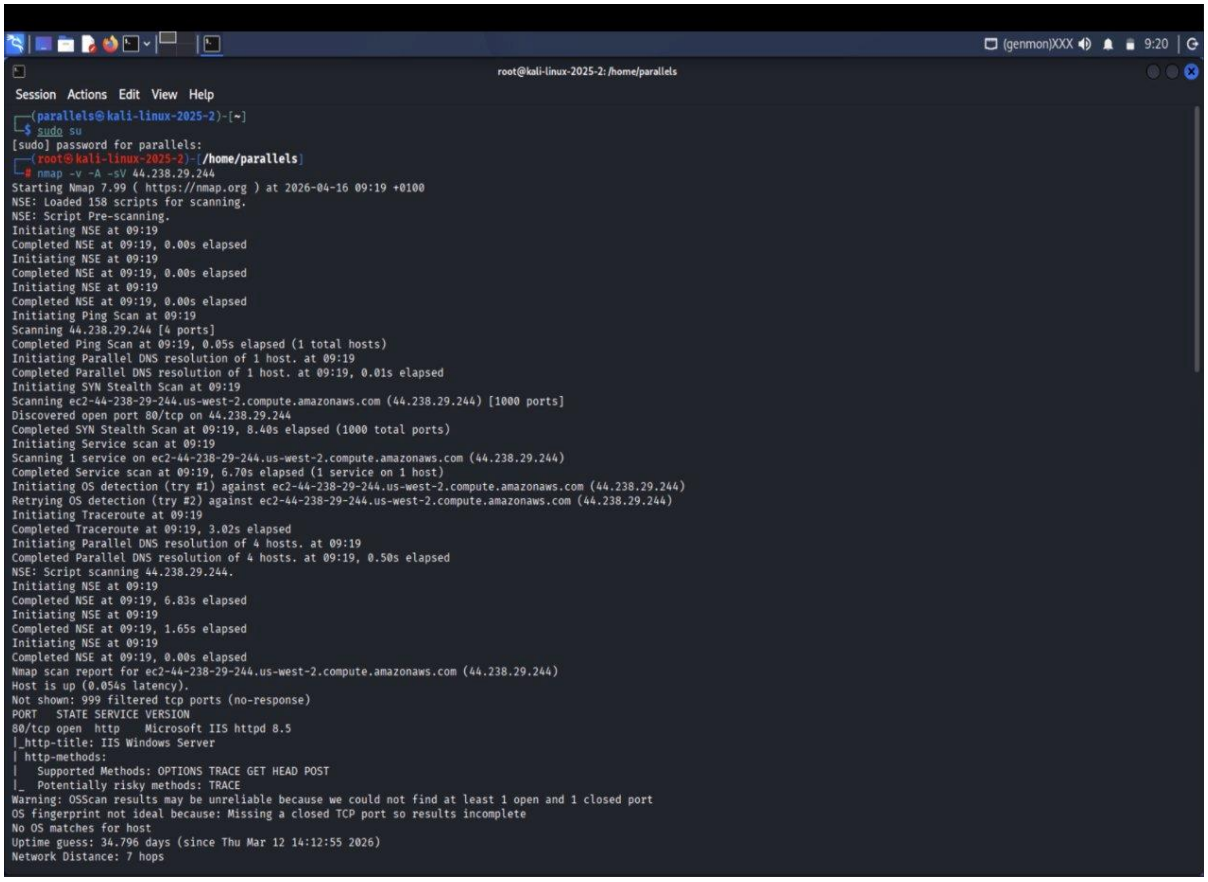
- Target website: http://testasp.vulnweb.com
- Target IP Address: 44.238.29.244
- Nmap Command: nmap -v -A -sV 44.238.29.244

Analysis

Port/Protocol	State	Service	Version
80/tcp	Open	HTTP	Microsoft IIS httpd 8.5

Port 80 hosts an active HTTP service running Microsoft IIS httpd 8.5. To reduce the risk of vulnerabilities, regularly applying the most recent security patches to web servers is crucial.

Step 3: Nmap Scan Screenshot — Fig(1)



Fig(1): Nmap Scan Result

Impact

Advantages:

1. Utilizing port 80 for HTTP traffic guarantees your website remains readily accessible, even for users who do not specify a port number in their browser.
2. Older browsers and some automated systems might rely on port 80 for accessing web resources.

Disadvantages:

3. Since HTTP traffic is unencrypted, it can be intercepted by malicious actors, leading to potential man-in-the-middle attacks. This can compromise sensitive data like login credentials and personal information.
4. Open ports can be exploited by attackers to gain unauthorized access to your server.

Mitigation Steps

5. Make sure you are using the most recent version of IIS/the web server. Regularly updating the software ensures you receive security patches that fix known vulnerabilities.
6. Redirect HTTP traffic to HTTPS (port 443) to ensure data encryption, providing a layer of protection against eavesdropping and man-in-the-middle attacks.
7. Implement logging and monitoring systems to quickly identify and respond to any suspicious activities.
8. Limit access to sensitive directories and files by configuring allow and deny directives in your server settings.
9. Conduct regular security audits and vulnerability scans to identify and address potential issues.

Resources Used: Kali Linux operating system, Network Mapping (Nmap) and Ping Tool.

[Task 2] — Brute Force Directories on http://testasp.vulnweb.com/

Attack Name: Brute Forcing Directories on Website

Severity: High (Score 7.0 – 8.9)

Reason: The severity is high because directories like admin, secured, vendor, and CVS often contain critical information and vulnerabilities that can be exploited to compromise the entire system.

Procedure Used

Still on Kali Linux, I performed a brute force attack on http://testasp.vulnweb.com using DIRB (DirBuster). My screenshot contains the directories that are present in the website.

Steps to Reproduce

Step 1: Target URL — http://testasp.vulnweb.com

Step 2: Brute Forcing URL — Using DIRB, the Kali Linux command-line tool, I scanned the URL to find all hidden directories. A strong directory wordlist was used for better results.

Results of the scan:

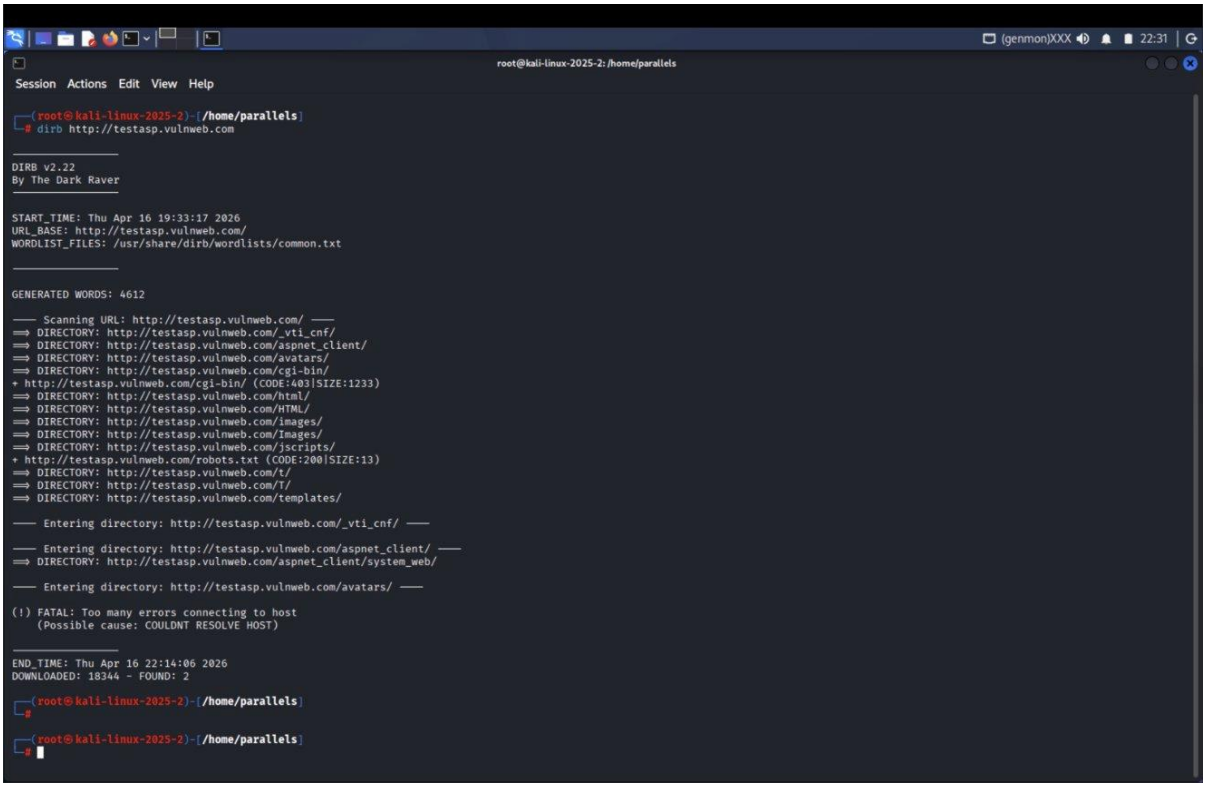
- Target website: http://testasp.vulnweb.com
- Command: dirb http://testasp.vulnweb.com
- Wordlist: /usr/share/dirb/wordlists/common.txt

Analysis — Discovered Directories

- http://testasp.vulnweb.com/_vti_cnf/
- http://testasp.vulnweb.com/aspnet_client/
- http://testasp.vulnweb.com/avatars/
- http://testasp.vulnweb.com/cgi-bin/ (CODE:403|SIZE:1233)
- http://testasp.vulnweb.com/html/
- http://testasp.vulnweb.com/HTML/
- http://testasp.vulnweb.com/images/
- http://testasp.vulnweb.com/Images/
- http://testasp.vulnweb.com/jscripts/
- http://testasp.vulnweb.com/robots.txt (CODE:200|SIZE:13)
- http://testasp.vulnweb.com/T/
- http://testasp.vulnweb.com/templates/

- http://testasp.vulnweb.com/aspnet_client/system_web/

Step 3: DIRB Scan Screenshot — Fig(2)



Fig(2): DIRB Scan Result

Impact

10. Directories such as these could contain configuration files, login credentials, or proprietary data. Accessing them might result in data breaches or unauthorized access.
11. These directories may contain files that can be targeted by attackers, enabling them to take advantage of these flaws to gain further access or execute harmful code.
12. Identifying and exploiting these directories can interrupt the website's regular functioning, leading to downtime or performance degradation.

Mitigation Steps

13. Limit access to important directories by implementing strong authentication and authorization measures. Make certain that only approved users are permitted to access these directories.
14. Regularly review and update configuration files to ensure they do not expose sensitive data. Remove default or unnecessary files that could be exploited.

15. Encrypt sensitive data in these directories to protect it from unauthorized access. Use strong encryption algorithms and secure key management practices.
16. Implement comprehensive monitoring and logging to detect and respond to unauthorized access attempts. Analyze logs regularly to identify suspicious activities.
17. Set appropriate file permissions to limit access to critical files. Ensure that sensitive files are not accessible to the public or unauthorized users.
18. Maintain regular backups of your data and have a well-defined recovery plan in place in case of a security incident.

Resources Used: Kali Linux operating system, DIRB tool.

[Task 3] — Intercept Login Credentials via Wireshark

Attack Name: Network Sniffing

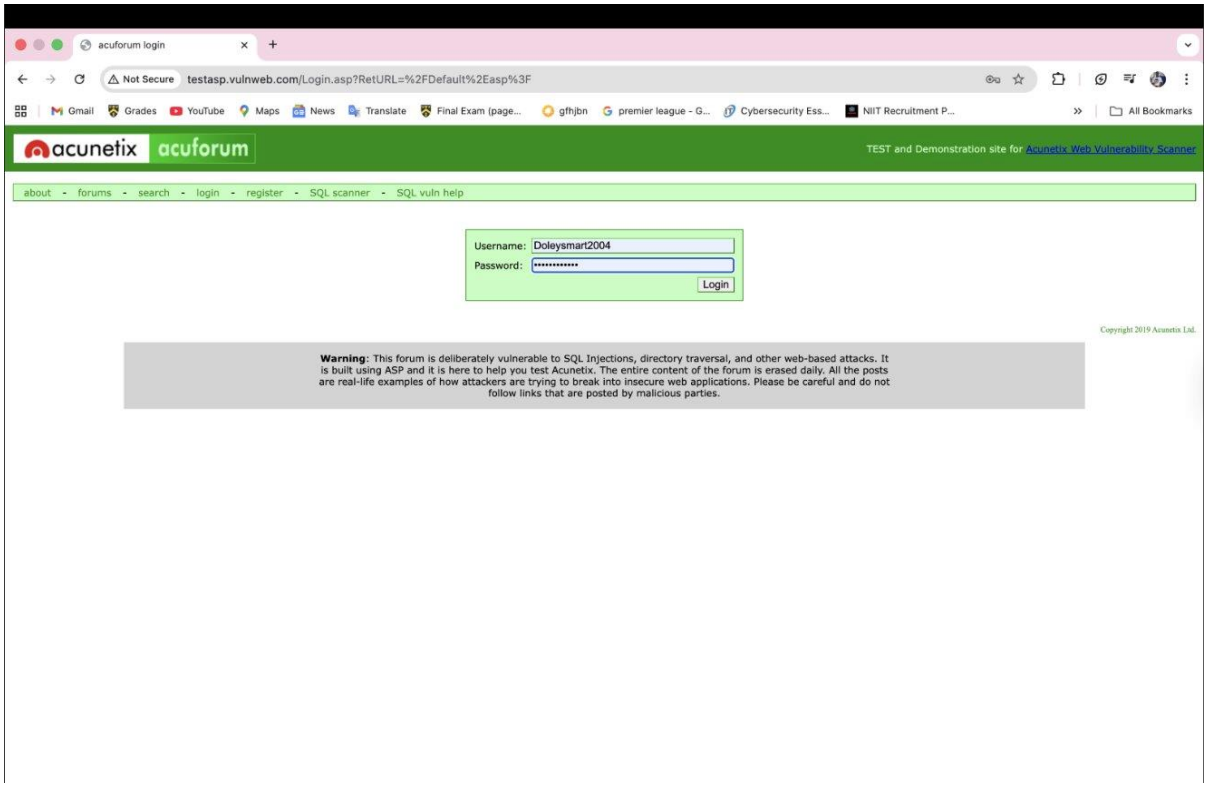
Severity: High (Score 7.0 – 8.9)

Reason: Network sniffing is highly severe because it can reveal sensitive information such as passwords and personal data, breach privacy, exploit network weaknesses, disrupt regular operations, and result in legal and compliance complications.

Procedure

19. Opened Wireshark on Kali Linux
20. Selected the active network interface
21. Started packet capture
22. Logged into the website: http://testasp.vulnweb.com
23. Applied Wireshark filter: http
24. Located the POST request related to login

Login Attempt Screenshot — Fig(3)



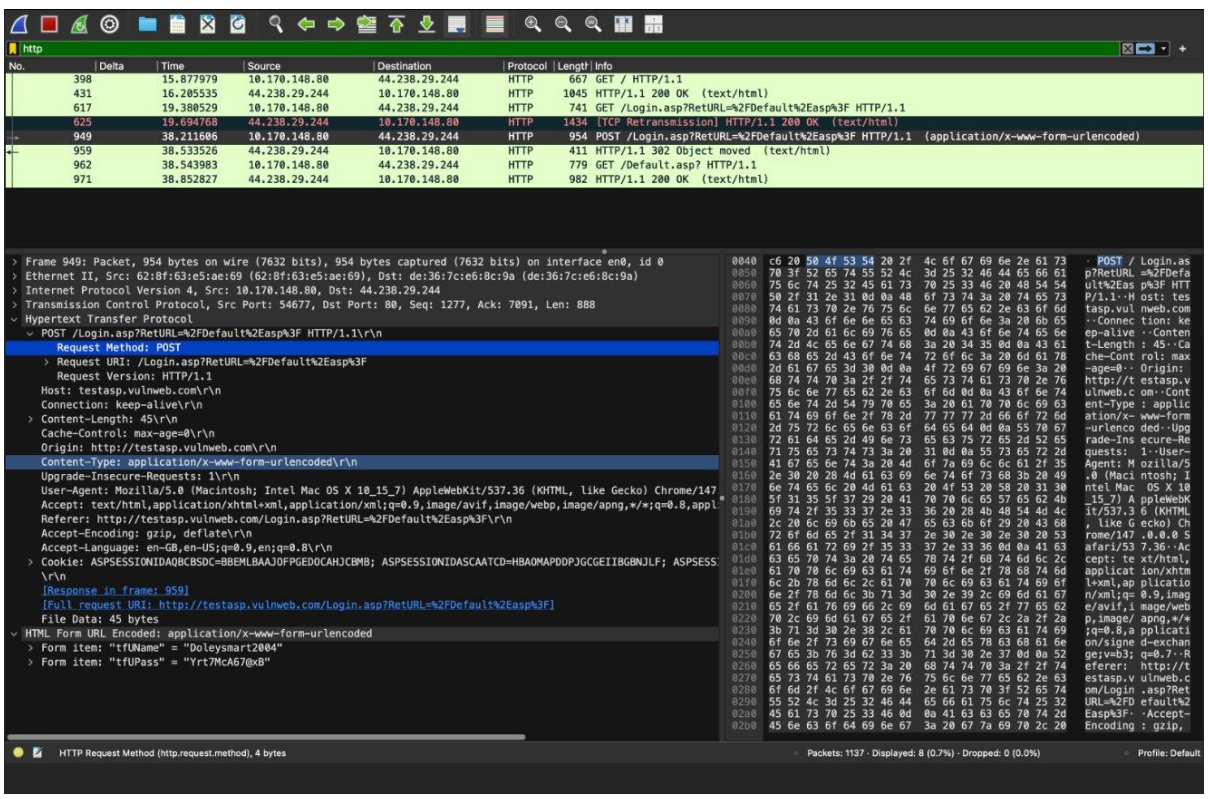
Fig(3): Login Attempt — acuforum login page (testasp.vulnweb.com)

Analysis

Wireshark captured numerous packets. By filtering for HTTP packets, the POST request containing the login credentials was isolated. The captured form data revealed the following credentials submitted in plaintext:

Field	Captured Value
Username (tfUName)	Doleysmart2004
Password (tfUPass)	Yrt7McA67@xB

Wireshark Credential Capture — Fig(4)



Fig(4): Wireshark Credential Capture — HTTP POST showing plaintext credentials

Impact

25. If login credentials are exposed in network traffic, unauthorized individuals could easily intercept and use them to gain access to user accounts.

26. If the credentials are linked to accounts with financial information, attackers could misuse them to commit financial fraud, including unauthorized transactions or accessing sensitive financial data.
27. Attackers can use the captured credentials to log in to user accounts, potentially leading to unauthorized access, data theft, and misuse of personal information.
28. Exposing login credentials puts user privacy at risk by giving attackers access to personal information, messages, and sensitive data connected to the accounts.

Mitigation Steps

29. Make certain that all communication between the client and server uses HTTPS/TLS encryption to ensure credentials are not sent in plaintext.
30. Adding an extra layer of security with Multi-Factor Authentication (MFA) can help protect accounts even if credentials are compromised.
31. Consistently monitor network traffic to identify any unusual activities or unauthorized access attempts. Use Intrusion Detection Systems (IDS) to help spot potential threats.
32. Keep all systems, software, and applications up to date with the latest security patches to prevent vulnerabilities that could be exploited.
33. Limit access to sensitive directories and data exclusively to authorized personnel by implementing strong authentication and authorization controls.

Resources Used: Kali Linux operating system, Wireshark tool.

Task Level (Intermediate)

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1	A file is encrypted using VeraCrypt. The password is encrypted in a hash format. Decode the password and enter it in VeraCrypt to unlock the file and find the secret code.	
2	Find the address of the entry point of the VeraCrypt executable using PE Explorer tool.	
3	Create a payload using Metasploit and make a reverse shell connection from a Windows 10 virtual machine.	
4	Make a deauth attack on your own network, capture the handshake, and crack the WiFi password using a custom wordlist.	

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[Intermediate Task 1] — Crack VeraCrypt Hash & Find Secret Code

Objective: Crack the hash in encoded.txt to obtain the VeraCrypt password, mount the encrypted container, and extract the secret code inside.

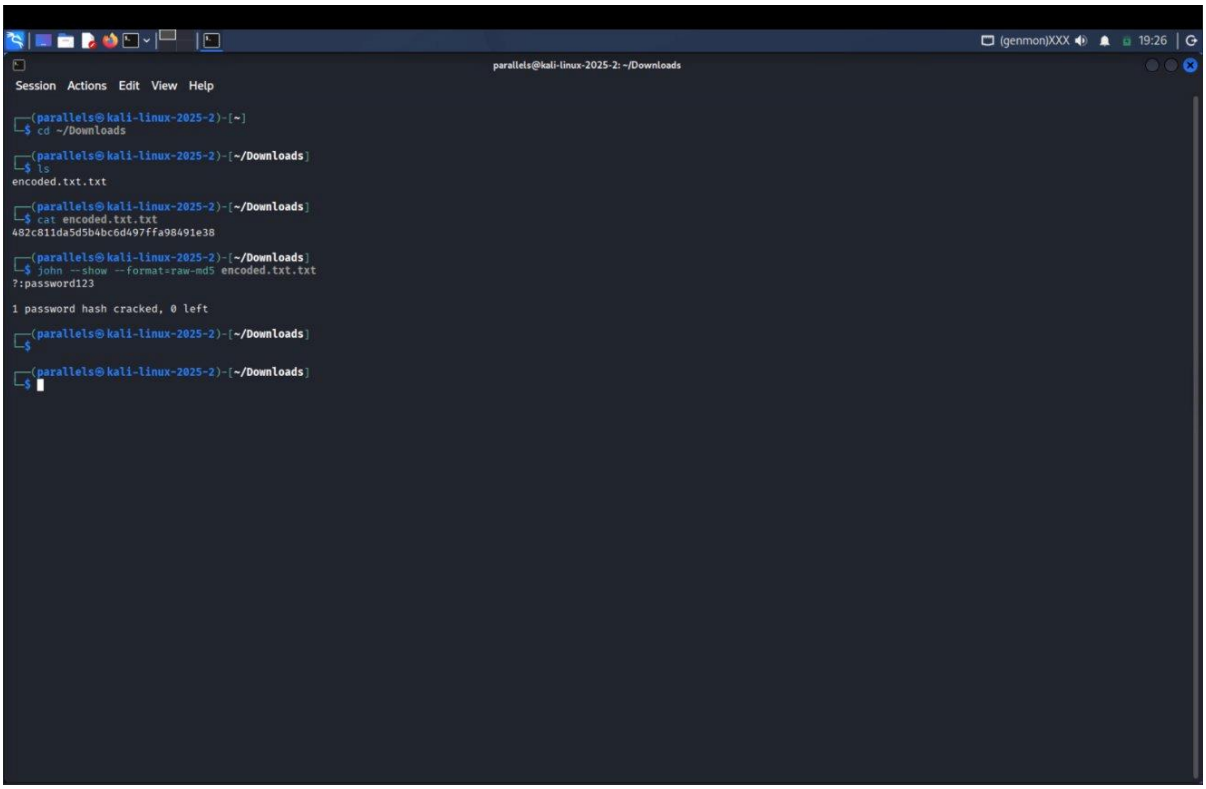
Resources Used

- Kali Linux (on Parallels Desktop)
- John the Ripper
- VeraCrypt

Procedure

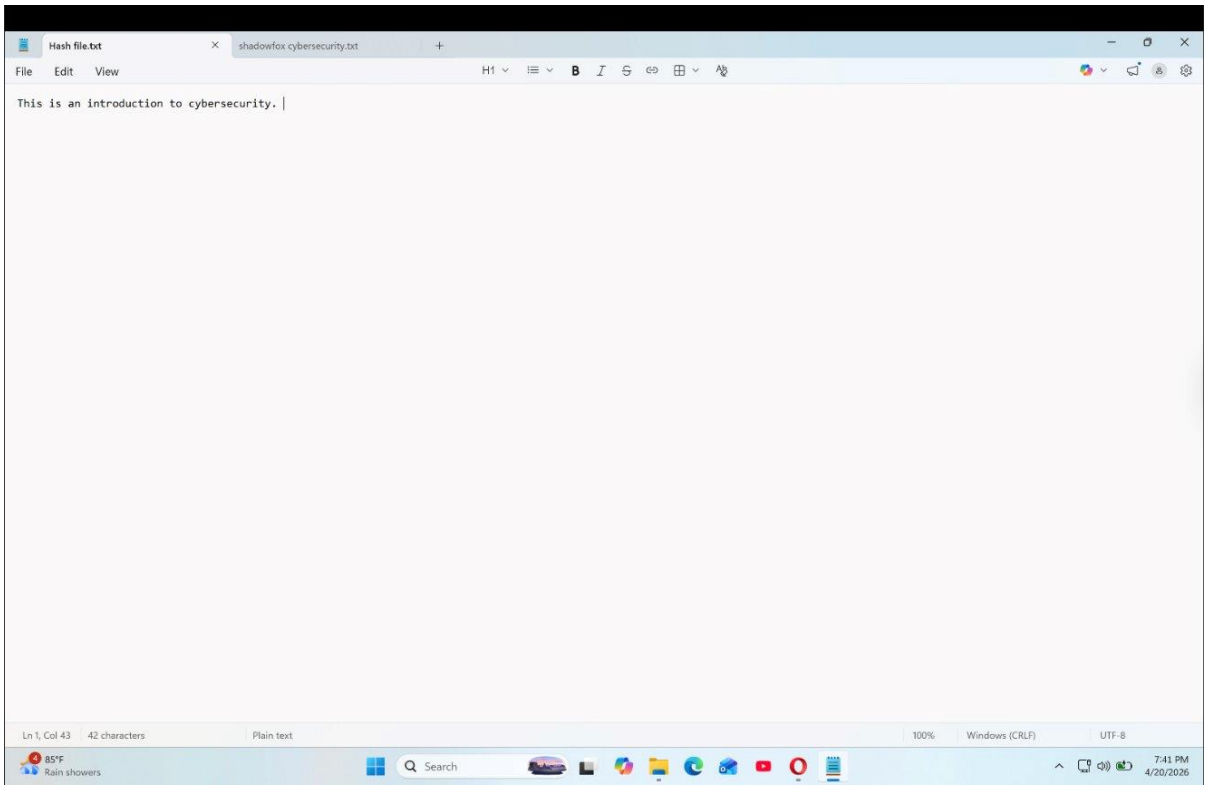
34. Navigated to ~/Downloads directory where encoded.txt.txt was located.
35. Ran cat encoded.txt.txt to view the hash: 482c811da5d5b4bc6d497ffa98491e38
36. Used John the Ripper with MD5 format to crack the hash: john --show --format=raw-md5 encoded.txt.txt
37. John the Ripper successfully cracked the hash and revealed the password: password123
38. Used the cracked password to mount the VeraCrypt encrypted container.
39. After mounting, opened the decrypted file and discovered the secret code.

John the Ripper — Hash Cracking — Fig(1)



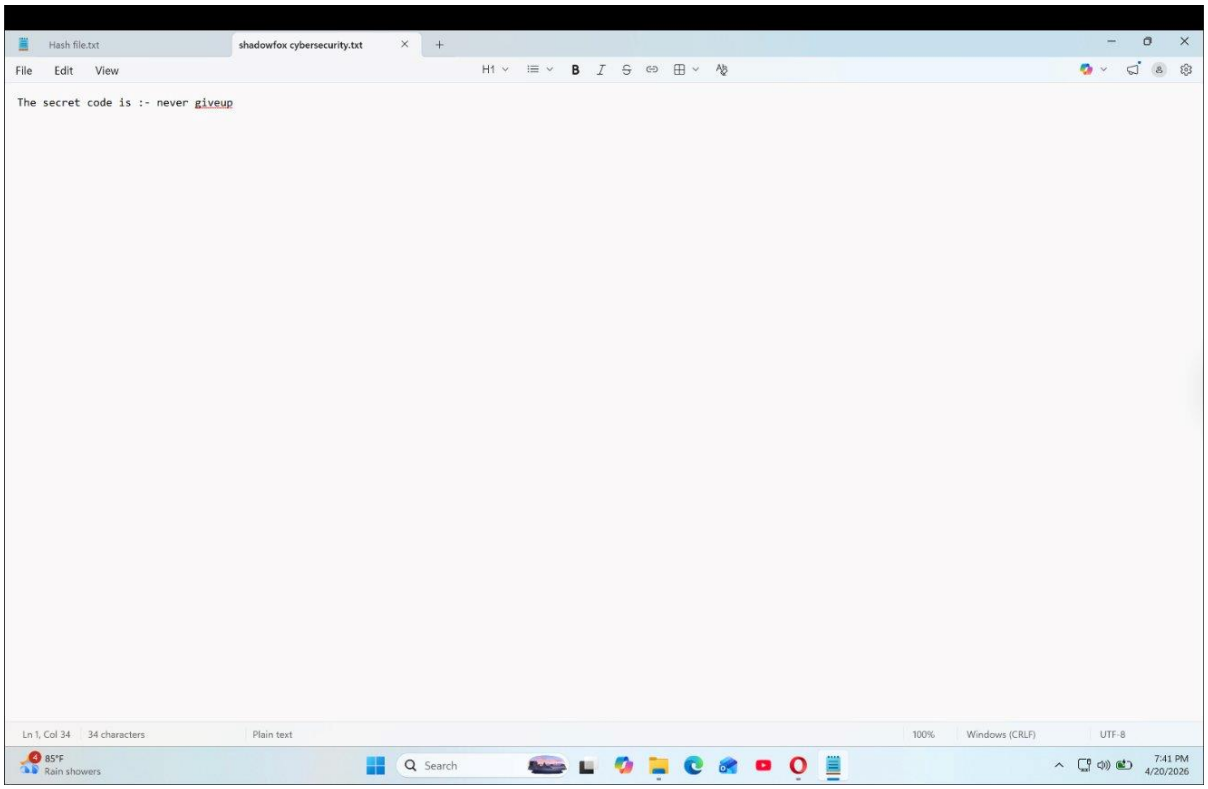
Fig(1): John the Ripper successfully cracking the MD5 hash — password: password123

VeraCrypt Screenshots — Fig(2)



Fig(2): VeraCrypt — Mounted encrypted container using cracked password

Decrypted Secret Code



Fig(2c): Secret code revealed: 'The secret code is :- never giveup'

The VeraCrypt encrypted container was successfully mounted using the cracked password. The secret code found inside the encrypted file was: The secret code is :- never giveup

[Intermediate Task 2] — Find Entry Point of VeraCrypt Executable via PE Explorer

Objective: Analyze the Portable Executable (PE) structure of the provided VeraCrypt executable file and extract the Address of Entry Point (EP) using PE Explorer.

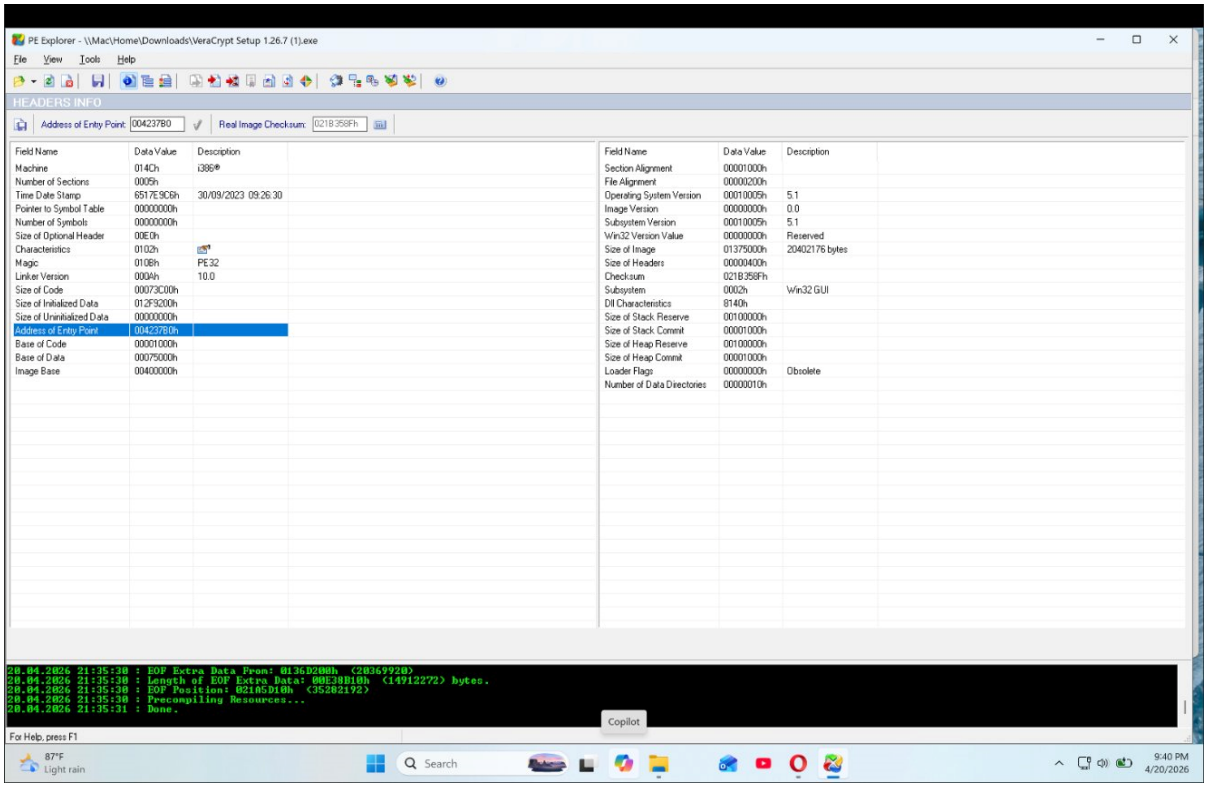
Resources Used

- Windows 10 Virtual Machine
- PE Explorer Tool (free PE file analyzer)
- VeraCrypt Setup executable (VeraCrypt Setup 1.26.7.exe)

Procedure

40. Opened PE Explorer on the Windows 10 Virtual Machine.
41. Loaded the VeraCrypt Setup 1.26.7.exe file into PE Explorer via File > Open.
42. Navigated to the Headers Info section within PE Explorer.
43. Located the 'Address of Entry Point' field in the PE header.
44. Recorded the displayed value.

PE Explorer Result — Fig(3)



Fig(3): PE Explorer — Address of Entry Point for VeraCrypt Setup 1.26.7.exe

Result

Field	Value
Address of Entry Point	**004237B0h**
Machine	014Ch (i386)
Magic / Format	010Bh / PE32
Subsystem	0002h (Win32 GUI)

The Address of Entry Point (EP) of the VeraCrypt Setup 1.26.7.exe executable is 004237B0h. This is the memory address where execution begins when the program is loaded into memory.